

Core Mathematical Formulas

python

```
import math
import numpy as np

# 1. Euclidean Distance (Distance between two points)
dist = math.dist([x1, y1], [x2, y2])

# 2. Quadratic Formula (Solving  $ax^2 + bx + c = 0$ )
d = (b**2) - (4*a*c)
sol1 = (-b - math.sqrt(d)) / (2*a)
sol2 = (-b + math.sqrt(d)) / (2*a)

# 3. Pearson Correlation Coefficient
correlation = np.corrcoef(x_list, y_list)[0, 1]

# 4. Standard Deviation
std_dev = np.std(data_list)

# 5. Sigmoid Function (Essential for Machine Learning)
def sigmoid(x):
    return 1 / (1 + math.exp(-x))
```

Text and String Operations

python

```
import re

# Clean whitespace and case
clean_text = raw_string.strip().lower()

# Regex: Extract all email addresses
emails = re.findall(r'[\w\.-]+@[\w\.-]+', text_body)

# F-Strings with formatting (2 decimal places)
pi_string = f"Pi is exactly {math.pi:.2f}"
```

Error Handling and Debugging

python

```
# Try-Except block with fallback
try:
    result = critical_value / user_input
except ZeroDivisionError:
    result = 0
except TypeError:
    result = float(user_input)
else:
    print("Success!")
finally:
    print("Execution finished.")
```

Essential Data Manipulation

python

```
# List Comprehension with conditional logic
filtered_data = [x**2 for x in raw_list if x > 0]

# Dictionary Comprehension (Merging two lists into a dict)
user_dict = {key: value for key, value in zip(keys_list, values_list)}

# Lambda Functions (Quick anonymous formulas)
area = lambda radius: math.pi * (radius ** 2)

# Sorting complex structures (Sort dict by values)
sorted_dict = dict(sorted(my_dict.items(), key=lambda item: item[1]))
```

File and Data I/O

python

```
import json
import csv

# Safe file opening (Auto-closes file)
with open("data.txt", "r", encoding="utf-8") as file:
    content = file.read()

# Fast JSON Reading
with open("config.json", "r") as jfile:
    data = json.load(jfile)

# Reading CSV rows cleanly
with open("dataset.csv", "r") as cfile:
    reader = csv.reader(cfile)
    header = next(reader) # Skips header
    rows = [row for row in reader]
```

Geometry Formulas

python

```
import math

# 1. Area of a Circle ( $A = \pi * r^2$ )
area_circle = math.pi * (radius ** 2)

# 2. Circumference of a Circle ( $C = 2 * \pi * r$ )
circumference = 2 * math.pi * radius

# 3. Area of a Triangle ( $A = 0.5 * b * h$ )
area_triangle = 0.5 * base * height

# 4. Area of a Trapezoid ( $A = 0.5 * (a + b) * h$ )
area_trapezoid = 0.5 * (side_a + side_b) * height

# 5. Volume of a Sphere ( $V = 4/3 * \pi * r^3$ )
volume_sphere = (4/3) * math.pi * (radius ** 3)

# 6. Pythagoras Theorem ( $c = \sqrt{a^2 + b^2}$ )
hypotenuse = math.sqrt(side_a**2 + side_b**2)
```

Use code with caution.

Algebra and Basic Operations

python

```
# 1. Slope of a Line ( $m = (y2 - y1) / (x2 - x1)$ )
slope = (y2 - y1) / (x2 - x1)

# 2. Linear Equation ( $y = mx + c$ )
y = (slope * x) + intercept

# 3. Midpoint of a Line
midpoint_x = (x1 + x2) / 2
midpoint_y = (y1 + y2) / 2

# 4. Percentage Calculation (What percent is part of whole?)
percentage = (part / whole) * 100

# 5. Percentage Increase/Decrease
percent_change = ((new_value - old_value) / old_value) * 100
```

Basic Statistics

python

```
# 1. Mean (Average)
mean = sum(numbers_list) / len(numbers_list)

# 2. Range (Max - Min)
data_range = max(numbers_list) - min(numbers_list)
```

Use code with caution.

Unit Conversions

python

```
# 1. Celsius to Fahrenheit
fahrenheit = (celsius * 9/5) + 32

# 2. Fahrenheit to Celsius
celsius = (fahrenheit - 32) * 5/9

# 3. Kilometers to Miles
miles = kilometers * 0.621371

# 4. Radians to Degrees
degrees = math.degrees(radians)
```